

Shafagh Rastegari

Bologna, Italy | shafaghrastegari@gmail.com | (+39)351 734 4192 | [linkedin.com/in/shafagh-rastegari-411767223](https://www.linkedin.com/in/shafagh-rastegari-411767223)
shafaghrastegari.github.io

About Me

Final-year AI Master student with a Computer Engineering background, specializing in Deep Learning, Machine Learning, NLP, and Large Language Models. Project experience includes building a production RAG system with AI agents and vector databases, fine-tuning models for NLP challenges achieving top-5 ranking, and developing an end-to-end LLM pipeline for bias mitigation. Seeking an AI/ML or software engineering role in production AI systems.

Education

University of Bologna	Bologna, Italy
Master of Science in Computer Science, Artificial Intelligence	Sep 2023 – Current
Iran University of Science and Technology	Tehran, Iran
Bachelor of Science in Computer Engineering	Sep 2017 – Oct 2022

Experience

Front-end Developer	June 2021 – March 2022
Targoman Intelligent Processing Company	Tehran, Iran
• Developed admin and user panel websites using Vue.js and TypeScript, integrating RESTful APIs for backend connectivity.	

Projects

ChatPaper — Agentic RAG Research Assistant <i>AI Agents, RAG, LLM</i>	Repository Link
• Built a RAG-powered assistant that lets researchers chat with academic papers using semantic retrieval, a LangGraph ReAct agent, and LLM answer generation, evaluated with RAGAS achieving 1.0 Faithfulness score.	
• Tools Used: Python, LangGraph, LlamaIndex, ChromaDB, Streamlit, RAGAS, OpenRouter API	
Bias Mitigation in Reasoning LLMs with Multi-Judge Pipeline <i>LLM, Reasonable AI</i>	Repository Link
• A pipeline to detect and remove biased reasoning steps of Chain-of-Thought in reasoning LLMs to reducing stereotype propagation without compromising accuracy, benchmarked model performance across bias score and accuracy on BBQ (Bias Benchmark for QA) and MBBQ (Multilingual BBQ) datasets.	
• Tools Used: Python, LangChain	
Multiple Couriers Planning Problem <i>Optimization</i>	Repository Link
• Implemented a Mixed-Integer Programming (MIP) using PuLP to efficiently solve courier planning problem, and Dockerize it as combinatorial decision making and optimization approach.	
• Containerized the solution via Docker, guaranteeing the execution across different systems.	
• Tools Used: Python, MIP, PuLP, Docker	
Subjectivity in News Articles <i>NLP, LLM</i>	Repository Link
• Fine-tune mDeBERTaV3-base model for 5 different languages in 3 different settings (Multilingual, Monolingual, and zero-shot) for a binary classification task in order to detect the Subjectivity or Objectivity of the news sentences a part of this challenge. Our results placed us among the top 5 teams in almost all languages.	

Technical Skills

Languages: Python, C++, Prolog, Vue.js, React, JavaScript, TypeScript

Libraries: LangChain, LangGraph, LlamaIndex, RAGAS, Streamlit, TensorFlow, Keras, scikit-learn, Pandas, NumPy, Matplotlib

Technologies: Git, Docker, Linux, ChromaDB, LaTeX

AI/ML: AI Agents, RAG Systems, Vector Databases, LLMs, Fine-tuning

Languages

Persian: native, **English:** proficient, **Italian:** beginner